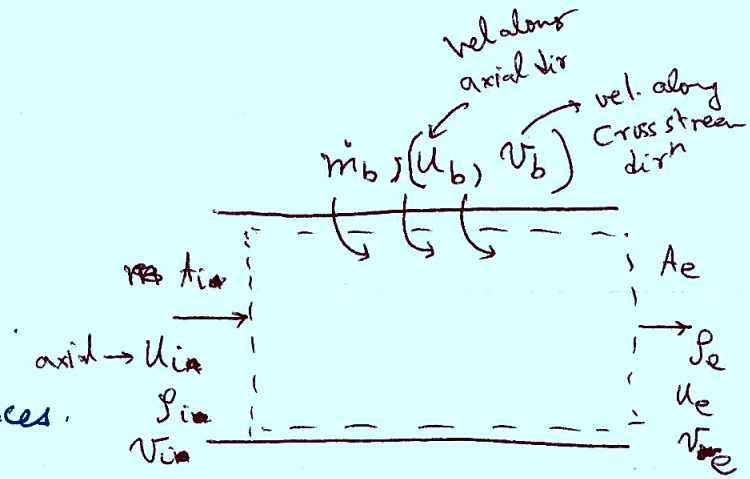
Lecture 6

Momentum Eqn:

$$\sum \bar{F} = \frac{d}{dt}\underbrace{(m\bar{V})}_{\text{mtm}}.$$



→ Forces represent production of mtm.

Forces can be divided into surface forces & body forces.

a) Surface forces : - normal (pressure)
- tangential (shear)

b) Body forces : - gravity
- magnetism
- electrical

- Tangential forces arise from viscosity

$$(\bar{\tau} = \bar{\nabla}\bar{V})$$

Viscosity is ~~[illegible]~~ & responsible for transport of mtm.

→ diffusive ~~[illegible]~~ (want to neglect)

→ inconsistent w/ neglecting diffusion, but will do it anyhow

- We will also neglected body forces

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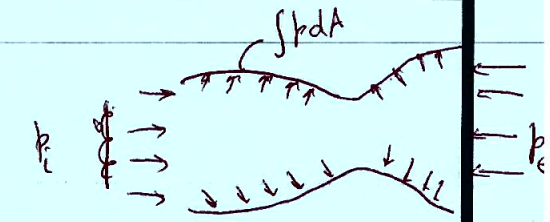
Cons. of mom. (for axial dir^n) (x, u)

$$\int_A u \rho u \, dA = \sum \int dF_x$$



Simple inlet & outlet, only pressure forces @ inlet & exit, & lumps all viscous & pressure forces along walls into $F_{b,x}$.

$$\boxed{A_i p_i + \rho_i u_i^2 A_i + \sum \dot{m}_b u_b + F_{bx} = A_e p_e + \rho_e u_e^2 A_e}$$

where $\sum \dot{m}_b u_b$ contribution from mass flowing thru boundaries.

Cons. of mtm in cross stream dir (y, v)

$$\boxed{\rho_i u_i v_i A_i + \sum \dot{m}_b v_b + F_{b,y} = \rho_e u_e v_e A_e}$$

~~Cons. of Energy.~~

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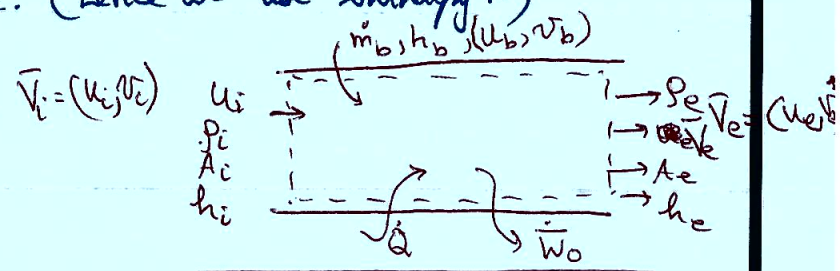
Cons. of Energy.

- Since system is flowing, there will be flow work. (Hence we use enthalpy h)
  (flow work $= p v = p/\rho$)
- Neglect potential energy
- Simple entrance & exit, 1D flow.



$$\boxed{\rho_i u_i A_i \left(h_i + \frac{u_i^2 + v_i^2}{2}\right) + \sum \dot{m}_b \left(h_b + \frac{u_b^2 + v_b^2}{2}\right) + \dot{Q} - \dot{W}_o = \rho_e u_e A_e \left(h_e + \frac{u_e^2 + v_e^2}{2}\right)}$$

$\dot{W}_o$: other forms of work besides flow work.

Eq^n of state

For ideal gas, $h = f(T)$ only; IGL $\Rightarrow$

$$\boxed{p = \rho R T}$$ (for non-ideal gas $h = f(T, p)$)

T/D 2nd Law : Not a conservation statement, but still needed.

$$ds \geq \frac{\delta Q}{T}$$

for reversible process, $ds = \dfrac{\delta q}{T}$.

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Start w/ 1st Law. for isolated system undergoing reversible process.

$$\delta q - \delta w = de + d(ke) + d(\underbrace{pe}_{\text{pot. energy}})$$

Neglect $d(ke)$ & $d(pe)$

$$\Rightarrow \quad \delta q + \delta w = de$$

Because the process is reversible.

$$T ds + \delta w = de$$

Limit δw to $p = \phi$ boundary work

$$\Rightarrow \delta w = -p dv$$

$$\Rightarrow \quad T ds - p dv = de$$

$$\Rightarrow \boxed{T ds = de + p dv}$$

But $h = e + pv$

$$dh = de + p dv + v dp$$

$$\Rightarrow \boxed{T ds = dh - v dp} \quad \text{Gibbs eq}^{n}.$$

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Speed of sound.

- A measure of how fast information propagates
- By defn

$$a^2 = \left.\frac{\partial p}{\partial \rho}\right|_{s=\phi} \quad \text{(isentropic : rev \& adia)}$$

$\rightarrow p : \rho R T$

$$\left.\frac{dp}{d\rho}\right|_{T=\phi} = RT$$ } for isothermal process. (assumed by Newton) (Wrong!! assumption)

$$p \sim \rho^{\gamma} \; ; \; p \sim T^{\gamma/\gamma-1}$$

$$\Rightarrow \boxed{a^2 = \gamma R T} \quad \text{for IG.}$$

For non-ideal gases (but still in LTE)

$$a^2 = \gamma R T \left\{ \frac{1 + \left.\frac{1}{p}\frac{\partial e}{\partial v}\right)_{T=\phi}}{1 - \rho \left(\frac{\partial h}{\partial p}\right)_{T=\phi}} \right\} \qquad \left(h, e = f(p, T)\right)$$

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→ Can introduce Mach #

$$M^2 = \frac{V^2}{a^2} = \frac{u^2+v^2}{a^2}$$

Stagnation prop. revisited (i.e. total properties)

- Use subscript 't'
- Using 1st law for adiabatic system, no work, no P.E.

$$h_t = h_0 + \frac{1}{2} V_0^2$$

For CPG, $h = c_p T$

$$\boxed{T_t = T_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)}$$

} subscript '0' ⇒ freestream cond'n

w/ 2nd law: isentropic process & CPG.

$$\boxed{p_t = p_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)^{\gamma/\gamma-1}}$$

$$\boxed{\rho_t = \rho_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)^{1/\gamma-1}}$$

} Isentropic process
Only valid for CPG.

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For non-isentropic process, but adiabatic process

$$T_t = \phi$$

$$p_t \neq \phi$$

$p_t \equiv$ a measure of an ability to do work. A loss of p_t due to irreversibilities degrades our ability to extract work from the system

$\Rightarrow$ want to minimize total pr. loss.

- All real processes are irreversible & entropy must increase
- Loss of stag pr. scales exponentially w/ entropy increase

$$\frac{p_{t2}}{p_{t1}} = \exp\left(\frac{S_1 - S_2}{R}\right) \quad , \quad S_2 \geq S_1$$

Shocks generate entropy $\rightarrow$ lose p_t

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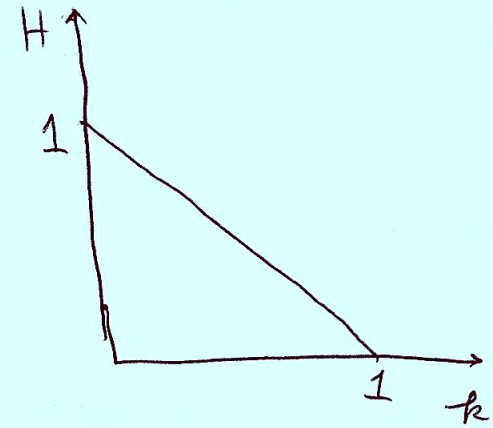
Recall an H-k diagram

$$H = \frac{C_pT}{C_pT_0}$$

$$k = \frac{V^2}{2C_pT_0}$$



1st Law w/ ~~&~~ no heat add^n yields

$$H + k = 1$$

We want to look @ family of lines/curves in H-k space that represent thrust.

→ Stream thrust funct^n, Sa

- Need heat add^n to generate thrust, but normalize everythink by T_{t_i}

Define impulse funct^n I,

$$I = pA(1+\gamma M^2) \qquad \{\text{Comes from } A(p+\rho u^2)\}$$

For 1D flow w/ single entrance & exit, we can show ~~eq~~ momentum eq^n (neglecting $\dot{m}_b$)

$$\underset{\text{Thrust}}{\tau} = \underbrace{\dot{m}(u_e-u_i)}_{\text{momentum thrust}} + \underbrace{p_eA_e - p_iA_i}_{\text{pressure thrust}}$$

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In general, we want momentum thrust & not pressure thrust for best fuel efficiency.

Look @ mom. flux term $\dot{m}u$

$$\dot{m}u = (\rho u A)\, u$$

$$= \frac{p}{RT} A u^2$$

$$= \frac{p}{RT} A M^2 a^2$$

$$= \frac{p}{RT} A M^2 \gamma R T$$

$$= \gamma p A M^2$$

Thus,

$$\tau = p_e A_e M_e^2 \gamma_e - p_i A_i M_i^2 \gamma_i + p_e A_e - p_i A_i$$

$$= \underbrace{p_e A_e (1 + \gamma M_e^2)}_{I_e} - \underbrace{p_i A_i (1 + \gamma M_i^2)}_{I_i} \qquad \{\text{assume } \gamma_e = \gamma_i = \gamma\}$$

$$\tau = I_e - I_i$$

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→ Specific thrust $= \dfrac{\mathcal{T}}{\dot{m}} = u_e - u_i + \dfrac{1}{\dot{m}}\left(p_e A_e - p_i A_i\right)$

$$= u_e - u_i + \frac{p_e A_e}{\rho_e u_e A_e} - \frac{p_i A_i}{\rho_i u_i A_i} \qquad \left(\text{recall, assumed } \sum \dot{m}_b = 0\right)$$

$$= u_e - u_i + \frac{RT_e}{u_e} - \frac{RT_i}{u_i}$$

$$= \underbrace{u_e\left(1 + \frac{RT_e}{u_e^2}\right)} - u_i\left(1 + \frac{RT_i}{u_i^2}\right)$$

Stream thrust funct^n

$$S_a = u\left(1 + \frac{RT}{u^2}\right) = \frac{I}{\dot{m}}$$

Specific thrust $\boxed{\dfrac{\mathcal{T}}{\dot{m}} = S_{a_e} - S_{a_i}}$

Units of S_a is $\left[\dfrac{m}{s}\right]$

⇒ Non-dimensionalize by $\left(C_p T_{t_i}\right)^{1/2}$ & define $\Phi \equiv \dfrac{S_a}{\sqrt{C_p T_{t_i}}}$ & use H&K diagram for representation

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HW1: Ch2: Prob 5, 6, 8, 10, 11

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